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Indian Institute of Information Technology Design and Manufacturing Kurnool

SyncSwitch – Smart Switch Board

DM104 Design Realization Practice

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SyncSwitch

Smart Control, Seamless Living

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PREFACE

The evolution of the modern smart home has introduced unprecedented convenience into daily life, yet it frequently overlooks two critical engineering requirements: operational redundancy and environmental safety.

The **SyncSwitch** project was conceptualized to address these specific vulnerabilities. Developing this system required navigating the complexities of non-blocking C++ logic, managing hybrid-cloud communications, and designing fail-safes for hardware-level interrupts.

This report documents the engineering journey, architectural decisions, and technical specifications of the **SyncSwitch v4.1.0**.

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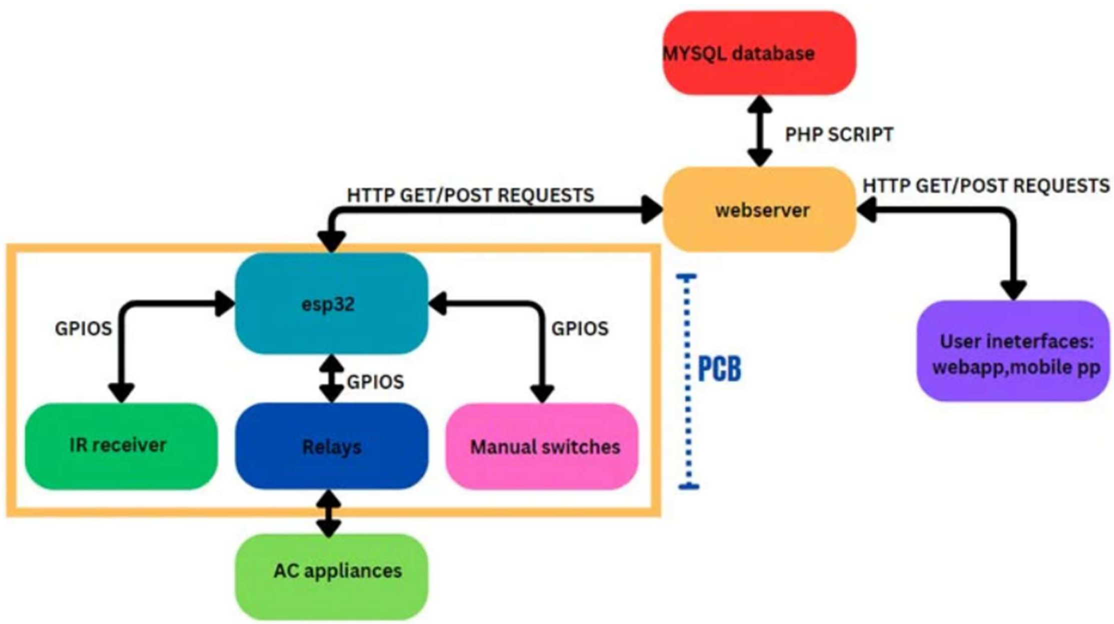
1. Product Overview

The **SyncSwitch v4.1.0** is an integrated Internet of Things (IoT) solution designed to bridge high-reliability home automation with environmental safety monitoring. The system addresses a common flaw in standard smart switches—single-point network failure—by providing a triple-redundant control interface (Physical, Infrared, and Cloud). Furthermore, it transitions from a simple automation tool to a home security device by integrating a continuous gas and fire monitoring system.

To ensure low latency while maintaining rigorous data analytics, the software architecture utilizes a hybrid-cloud approach: Sinric Pro handles user control and WebSocket communication, while a Firebase Realtime Database operates as an asynchronous background event logger.



2. Hardware Architecture & Components

2.1 Core Microcontroller: ESP32

- **Architecture:** Dual-core 32-bit Xtensa LX6 microprocessor running at 240 MHz.
- **Connectivity:** Integrated 802.11 b/g/n Wi-Fi and Dual-mode Bluetooth.
- **Operating Voltage:** 3.3V logic level.
- **Thermal Tolerance:** -40°C to +85°C.
- **Engineering Role:** Acts as the central processing unit. Its dual-core nature allows it to manage the complex Wi-Fi stack and secure WebSocket connections on one core, while simultaneously polling sensors and decoding IR signals on the other, ensuring zero perceived lag.

2.2 Environmental Safety Sensor: MQ-2 Gas & Smoke Module

- **Detection Range:** 300 to 10000 ppm. Highly sensitive to Propane, Hydrogen, LPG, and general combustible smoke.
- **Hardware Implementation:** Connected to the ESP32's 12-bit Analog-to-Digital Converter (ADC), providing resolution values from 0 to 4095.
- **Calibration Trigger:** The system is hard-coded with a FIRE_THRESHOLD of 4100.
- **Durability:** Features a long-life electrochemical sensor with an internal SnO_2 heating element to maintain accuracy across fluctuating indoor humidity levels.

2.3 Local Wireless Control: TSOP Infrared Receiver (38kHz)

- **Frequency:** Tuned specifically to 38kHz to match standard household remote controls.
- **Signal Integrity:** Features built-in PCM frequency filtering and internal metal shielding. This prevents Electromagnetic Interference (EMI) from the ESP32’s

onboard Wi-Fi antenna from corrupting the incoming infrared data streams.

2.4 Power Switching: 3-Channel Relay Module

- **Electrical Rating:** 10A at 250V AC / 10A at 30V DC per channel.
- **Circuit Protection:** Utilizes opto-isolated inputs. This critical safety feature physically separates the high-voltage AC side from the 3.3V DC logic side using light signals, protecting the ESP32 from fatal back-EMF spikes during switching.
- **Mechanical Durability:** Rated for up to 10 million mechanical switching cycles.

2.5 ESP32 Pin Configuration

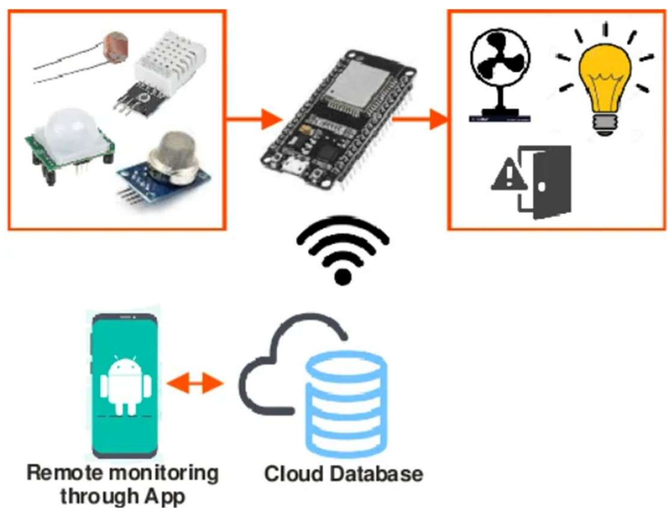
Component	ESP32 Pin
IR Receiver	GPIO 15
Buzzer	GPIO 13
Gas Sensor	GPIO 34
Relay 1	GPIO 2
Relay 2	GPIO 4
Relay 3	GPIO 5
Switches	14, 27, 26

```
const int IR_RECEIVE_PIN = 15;
const int BUZZER_PIN     = 13;
const int GAS_SENSOR_PIN = 34;

const int LED_PINS[]     = {2, 4, 5};
const int WALL_SWITCH_PINS[] = {14, 27, 26};
```

3. Software Architecture & Hybrid Cloud Integration

The SyncSwitch operates on a highly optimized C++ codebase utilizing the Arduino framework. It departs from traditional monolithic IoT structures by splitting its cloud responsibilities.



3.1 Primary Control Layer (Sinric Pro)

- **Function:** Serves as the consumer-facing user interface (Mobile Dashboard) and handles voice assistant integration (Alexa/Google Home).
- **Protocol:** Uses secure WebSockets to maintain a persistent, low-latency connection to the device, allowing relay states to change in under 500ms when triggered remotely.

3.2 Analytics & Auditing Layer (Firebase RTDB)

- **Function:** Operates as a "Black Box" flight recorder for the switchboard.
- **Implementation:** Uses RESTful API pushes to send data asynchronously. It logs network connection events, physical button presses, and real-time gas sensor readings without slowing down the primary control loop.

- **Time Synchronization:** Utilizes a Network Time Protocol (NTP) server offset to Indian Standard Time (IST) to ensure all logs have accurate timestamps.

3.3 Local Concurrency & State Machine

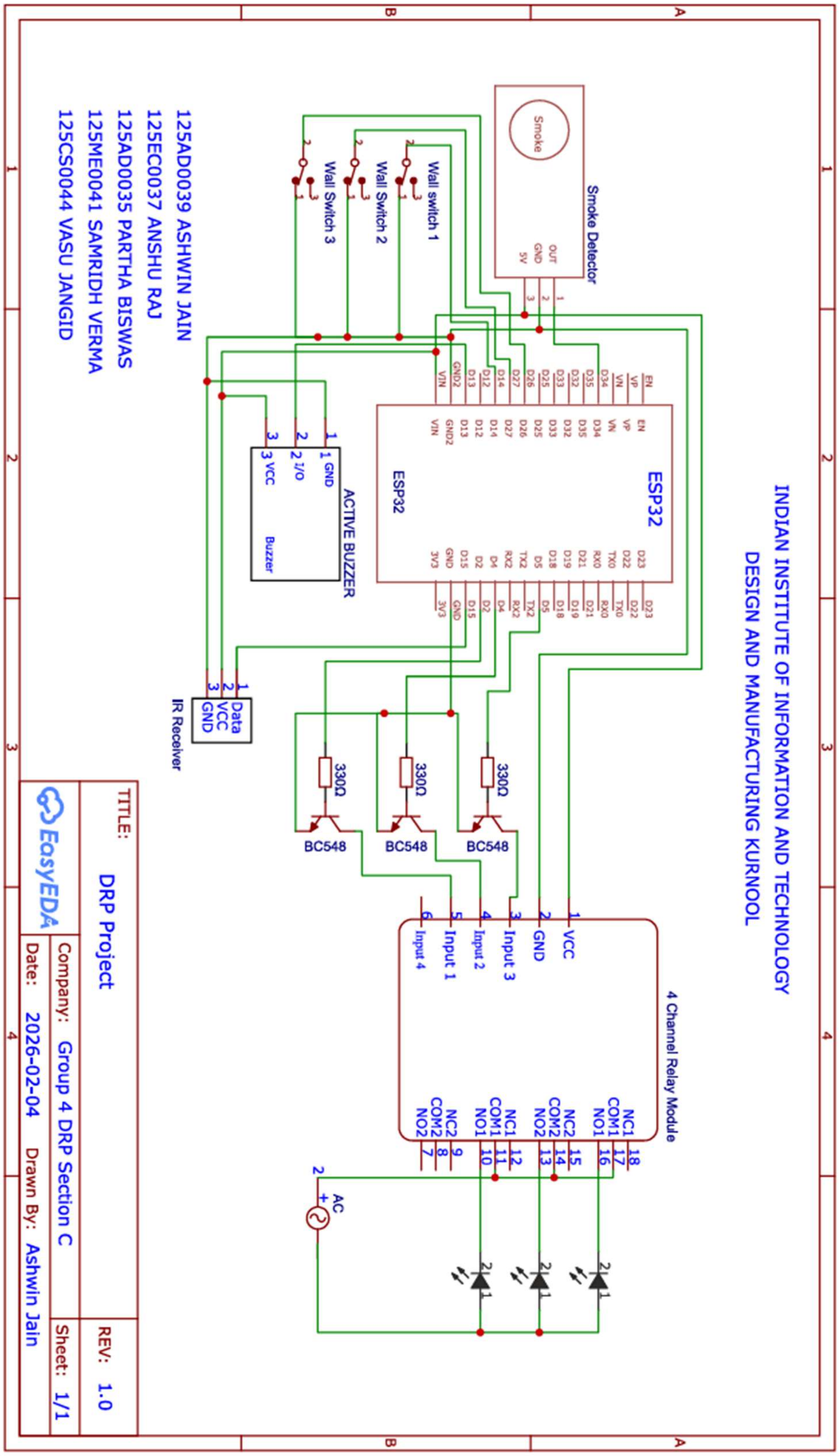
- **Non-Blocking Logic:** The codebase strictly avoids delay() functions. Instead, it utilizes millis() tracking to manage the buzzer duration and the 5000ms gas sensor polling interval.
- **State Synchronization:** A global array (ledStates[]) acts as the single source of truth. If a relay is toggled via an IR remote, the state machine instantly propagates this change to the Sinric Pro dashboard and pushes an event log to Firebase.

4. Technical Specifications & Operating Parameters

4.1 Specification Table

Specification	Value / Description
Input Power Source	5V DC (via regulated power supply)
Maximum Switching Load	15A Total (across all 3 channels)
Local Response Latency	< 150ms (Physical Switch / IR Remote)
Cloud Response Latency	< 500ms (Sinric Pro Dashboard)
Operating Humidity	< 90% Relative Humidity (Non-condensing)
Network Protocol	WPA/WPA2 Personal (2.4 GHz Band Only)
Audible Alert Output	Active High Piezo Buzzer (Software-driven pulse)

4.2 Schematic Diagram of SyncSwitch



5. System Operations & User Manual

5.1 Initial Boot & Calibration

1. Ensure the ESP32 and Relay Module are securely connected to the 5V power supply.
2. Upon power-up, the MQ-2 sensor requires a 60-second "pre-heat" phase for the internal element to stabilize.
3. The device will automatically scan for the pre-configured Wi-Fi network. A successful connection will trigger a 100ms audible beep and log a "Device Booted" event in the Firebase database.

5.2 Device Control Modalities

- **Mobile/Voice:** Open the Sinric Pro application. Toggling any of the three virtual switches will instantly actuate the corresponding physical relay and emit a short confirmation beep.
- **Infrared Remote:** Point a standard IR remote at the TSOP receiver. Programmed hex codes (e.g., Master On, Master Off, or individual toggles) will bypass the network and switch the relays directly.
- **Wall Switch:** Physically flipping the connected manual wall switch will trigger a state-change algorithm (ignoring mechanical bounce) to toggle the relay.

```
void toggleLED(int index, SinricProSwitch& mySwitch) {
    ledStates[index] = !ledStates[index];
    digitalWrite(LED_PINS[index], ledStates[index] ? HIGH : LOW);
    beep(100);
}
```

codesnap.dev

5.3 Safety & Emergency Protocols

- The system monitors ambient air quality every 5 seconds.

- If the analog reading surpasses the safe threshold (4100), the device enters **Alarm State**, triggering a continuous pulsing sequence from the hardware buzzer.
- System administrators can log into the Firebase console to review the exact timestamp and severity of the gas spike.

6. Maintenance & Troubleshooting

- **Offline Behaviour:** If the Wi-Fi status drops, the device intelligently suspends cloud communication attempts to prevent processor hanging. Physical and IR control will continue to operate with zero latency.
- **Relay Degradation:** To maximize the lifespan of the mechanical relays, avoid rapid, successive switching (flickering) of heavy AC loads.
- **Sensor Cleaning:** Ensure the mesh covering the MQ-2 sensor remains free of heavy dust accumulation to maintain accurate environmental readings.

7. Conclusion

7.1 Future Scope

- Development of a dedicated mobile application for enhanced user interface and control.
- Integration of AI/ML models for predictive gas detection and anomaly analysis.
- Implementation of a custom PCB design to improve compactness and reliability.
- Addition of more sensors (temperature, humidity, motion) for extended smart home capabilities.
- Integration with advanced voice assistants and home ecosystems.
- Enhancement of security features such as encrypted communication and user authentication.
- **Online code version update (OTA)** to allow remote firmware upgrades without physical access to the device.

7.2 Abbreviations

Abbreviation	Full Form
IoT	Internet of Things
ESP32	Microcontroller with Wi-Fi and Bluetooth capabilities
ADC	Analog to Digital Converter
GPIO	General Purpose Input Output
IR	Infrared
MQ-2	Gas and Smoke Sensor Module
RTDB	Realtime Database
NTP	Network Time Protocol
API	Application Programming Interface
EMI	Electromagnetic Interference
AC	Alternating Current
DC	Direct Current

7.3 Conclusion

The SyncSwitch v4.1.0 provides a reliable and efficient smart home solution by combining automation with safety monitoring. Its triple control system ensures uninterrupted operation, while the ESP32 enables fast, real-time processing. The integration of gas detection and hybrid cloud architecture makes the system practical, scalable, and suitable for real-world applications.

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